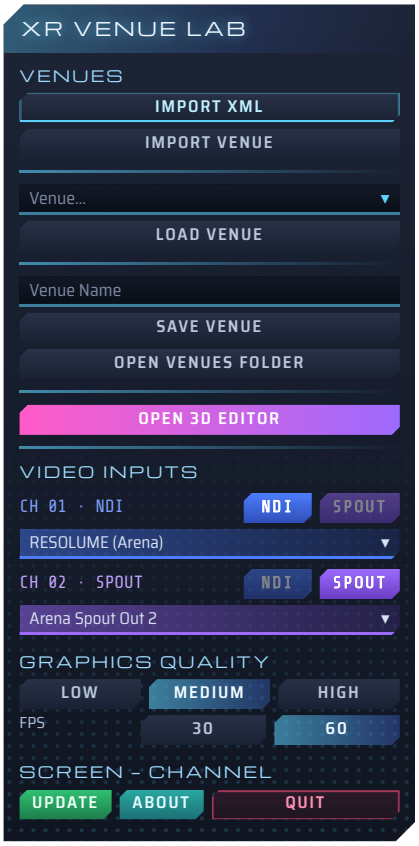
XR Venue Lab – Quick Guide

VERSION 1.0 · 11/03/2026 · UNREAL ENGINE 5.8

Buttons in this guide have the same shape and color as in the program: NORMAL ACTIVE MAIN EDITOR REVERT / RESET QUIT / CLOSE UPDATE ABOUT. In VIDEO INPUTS, NDI is always blue and SPOUT always violet. The selection is shown in magenta. Keys look like this: F.



Main menu (reference)

Moving in Unreal (technician)

You move as the technician, walking or flying.

KEY	WHAT IT DOES
W A S D	Walk (or fly) forward, left, back and right.
Mouse	Look around.
Shift	Toggles fast mode (3 times faster): one tap turns it on, another turns it off.
F	Toggles fly mode: you move where you look. Press F again to land.
Q / E	While flying: Q goes up, E goes down.
R	Takes you back to the front of the room, facing the screens.
C	Opens or closes the CAMERAS panel (see Cameras).
O	Opens or closes the REC OUTPUT panel, below CAMERAS (see Recording).
T or 0 · 1 … 7	T (or 0) goes back to the technician; 1 to 7 go straight to each camera. Pressing the same number again changes its view.
Space	Open or close the 3D editor (on exit you go back the way you were).
ESC	Open or close the main menu.

In fly mode you collide with walls, ceiling and floor. When you open the 3D editor the technician position is saved and restored when you close it (if it ended up outside the room, it reappears in front, facing the screens). The technician view is 1.65 m high, at the eye level of a standing person.

Main menu (ESC)

From the menu you create, load and save venues, and choose which video goes into each channel.

Venues

BUTTON	WHAT IT DOES
IMPORT XML	Creates a new venue from the Resolume XML (Advanced Output) and opens the 3D editor so you can arrange the screens.
IMPORT VENUE	Opens a venue (.json) from any folder, for example one a colleague sent you.
Dropdown + LOAD VENUE	Pick a saved venue and load it (with its groups and magnet attachments).
Venue Name + SAVE VENUE	Saves the venue with that name. If it already exists, it asks: OVERWRITE replaces it and CANCEL goes back without saving.
OPEN VENUES FOLDER	Opens the venues folder in Windows Explorer. To share a venue, send its .json file.
OPEN 3D EDITOR	Enters edit mode; also with Space (see below).
QUIT	Exits the program (bottom right; also the window X or Alt + F4). If the venue has unsaved changes it warns you first: QUIT WITHOUT SAVING or GO BACK. The same warning appears before IMPORT XML, IMPORT VENUE and LOAD VENUE, with CONTINUE WITHOUT SAVING or GO BACK.
ABOUT	Between UPDATE and QUIT. Opens the credits window: version, author, contact and the third-party technology the program uses (Unreal Engine, NDI®, Spout and the fonts), plus an invitation to share what you make and tag XR Venue Lab on social media. YOUTUBE opens our channel (youtube.com/@XRVenueLab) and CLOSE closes it.
UPDATE	Bottom left. Checks for a new version (see UPDATE below).

VIDEO INPUTS (4 channels)

- Each channel (CH 01 to CH 04) has NDI and SPOUT buttons; the active type is lit. The channel name and its dropdown take the same color (blue NDI, violet SPOUT), so you can tell each type at a glance.
- Below, the dropdown lists the sources of that type: with NDI, every source on the network; with SPOUT, the senders on this PC. The Spout list refreshes every time you open it.
- Changes apply instantly, with no confirm button.
- SPOUT tip: for better performance, enable Frame count in SpoutSettings (the official Spout tool) and restart Resolume. XR Venue Lab then copies only new frames; it also works without it, but copies every frame.

GRAPHICS QUALITY

Screen video looks the same at all three levels: only the room lighting changes.

BUTTON	WHAT IT DOES
LOW	No light bounces or reflections; screens do not light the room. The lightest.
MEDIUM	Medium-quality bounces and reflections; screens light the floor and walls. Recommended.
HIGH	Full bounces and reflections and higher-quality shadows. The heaviest.

The active level stays highlighted and is remembered next time (the first time it starts at MEDIUM). Below, FPS sets the frame-rate cap, independent of the level: 30 (frees the GPU for Resolume or other video software on the same PC, and lowers camera lag when the GPU is busy) or 60 (default, smoothest movement). It is also remembered.

SCREEN – CHANNEL

One row per Resolume Screen: choose which channel (CH 01 to CH 04) shows that group of screens. Applies instantly.

UPDATE

When it starts, the program checks on its own whether there is a new version. If there is, a green dot blinks softly next to UPDATE. Clicking it opens a window with the list of improvements: UPDATE opens the installer download page and CANCEL closes it. If you are up to date, the window says UP TO DATE. It needs internet: offline it tells you it could not check, and the program keeps working as usual. To install a new version, close XR Venue Lab before running the installer; your venues are not touched.

Cameras

Besides the technician view there are 7 cameras that place themselves from the room size and where the screens are (they recalculate when you change venue or XML). While you look through a camera the technician does not move: when you go back with T it is where you left it. When you change camera or view, a small caption appears at the bottom (for example CAM 6 · ORBIT · MEDIUM) and fades after 3 seconds.

KEY	WHAT IT DOES
1 FOH	From the back of the audience, centered, looking at the screens. Views: FLOOR (at floor level) · EYES (1.65 m) · MID (half the height) · HIGH (under the ceiling).
2 X-RAY	360° turn outside the room, looking at the center from above, always in X-RAY view (see below): the whole venue from a distance. Speed SLOW · MEDIUM · FAST. Distance and height are calculated from the room size and can be adjusted in the panel.
3 LEFT	Back left side, diagonal toward the screens. Same 4 heights as FOH.
4 RIGHT	Back right side. Same 4 heights as FOH.
5 STAGE	On stage. AUDIENCE: like the DJ, looking ahead at the audience. SCREENS: from the stage edge, looking at the stage screens.
6 ORBIT	Moving camera around the room, always looking at the screens. Speed SLOW · MEDIUM · FAST and mode SWING (arc back and forth in front) or endless full turn LOOP CW / LOOP CCW.
7 TRAVEL	Straight dolly from the back of the audience toward the stage and back, always facing forward. Speed SLOW · MEDIUM · FAST.

> The view chosen for each camera is remembered: if you are on the technician and press 1, it shows the last height you picked. The stage height is not known yet, so STAGE assumes a 1.2 m stage.

CAMERAS panel (C)

BUTTON	WHAT IT DOES
TECH (T) FOH (1) … TRAVEL (7)	Shows that camera; the active one is lit. Its key is in brackets. With the panel open the mouse is free and the keys still work. The panel sits against the right edge of the screen.
SCREEN X-RAY	Puts the screen in X-RAY view with any camera (also with AUTO SWITCH). Since PGM is a copy of the screen, it goes out that way too. With camera 2 it is always on.
VIEW / SPEED	The views (or speeds) of the camera you are looking through.
FOV	Lens opening of the current view (the technician too), 60 to 120 in steps of 5; only the number is shown. AUTO lit = automatic value (on TECH, the original opening).

X-RAY: DISTANCE, HEIGHT	Camera 2: how far from the center it turns (1 m steps; farther or closer) and at what height. An A next to the value means it is automatic.
ORBIT: MODE, RADIUS, HEIGHT, ARC	Movement mode, how far from the center it orbits, at what height and how wide the arc is (ARC only in SWING).
TRAVEL: START, END, HEIGHT	Where it starts (meters from the back of the audience), where it ends (meters before the stage screens) and at what height.
AUTO (path)	Sets the path of that camera back to the automatic one for the room.

AUTO SWITCH

Builds a "program" feed that switches cameras by itself, ideal for recording a single output. Each step is a camera with its view and seconds: + ADD CURRENT VIEW adds what you are seeing (8 s), – / + changes the seconds, ▲ / ▼ changes the order and R removes the step. START starts it (the step on air is shown in red and the caption at the bottom is hidden, to keep the recording clean); STOP or any camera key stops it.

PRESETS

Cameras do not depend on the venue: views, FOV, paths and the AUTO SWITCH sequence are remembered when you close the program. You can also save them with a name (SAVE) and load them with any venue (click the name). R deletes a preset (asks to confirm with SURE?).

X-RAY view

Shows the venue the way the 3D editor does, a kind of "X-ray": the room as a see-through box, the floor grid and the disabled screens in their ghost color, without slice names. It follows what you chose in the editor: GRID (and its STEP) and DISABLED: GHOST / INVISIBLE. If the editor has not been opened yet, it uses a 0.5 m grid and ghosts.

- On screen: the X-RAY button in the SCREEN row of the CAMERAS panel (works with any camera and with AUTO SWITCH), or camera 2, which is always in X-RAY.
- On recording outputs: the X-RAY button of each output (REC OUTPUT). PGM follows the screen.
- It does not touch the real room or the editor: each view chooses whether it sees the normal room or the X-RAY copy. While nothing uses X-RAY, it costs nothing.

Recording: REC OUTPUT (0)

Up to 4 video outputs to record or send to another program (OBS, Resolume, vMix...). Each one carries the screen (PGM), the technician view or any camera, via Spout (same PC) or NDI (another PC on the network). Panels, captions and buttons never show on the outputs: only the 3D image, identical to the screen. Outputs always start off when the program opens; what you chose for each one is remembered.

BUTTON	WHAT IT DOES
XRVL OUT 1 ... 4	Each output title = the name you will find it under in OBS or Resolume. Its color shows the mode (blue NDI, violet SPOUT); on the right, the frames per second it is really sending. When the output is on, its box has a red border.
ON / OFF	Turns that output on or off.
NDI / SPOUT	Where it goes. SPOUT: to another program on the same PC (fastest). NDI: to another PC on the network (uses more CPU; with nobody connected it does not spend on rendering).
PGM TECH 1 ... 7	What goes to that output. PGM is the screen as it is (AUTO SWITCH or the camera you pick by hand). Only one output can be PGM. Each camera view (height, speed, FOV, path) is chosen in the CAMERAS panel and the output follows it live.
1080 / 720	Resolution: 1920 × 1080 or 1280 × 720. One click switches.

30 FPS / 60 FPS	Frames per second of that output.
HIGH / LIGHT	HIGH: identical to the screen (Lumen lighting and reflections). LIGHT: much lighter, no Lumen, reflections or dynamic shadows (looks different). Not used on PGM.
X-RAY	That output in X-RAY view. On PGM it follows the screen; with camera 2 it is always on.
ALL OFF · CLOSE (O)	Turns every output off · hides the panel (outputs keep running).

> Performance: PGM costs almost nothing (it copies the screen). Each camera in HIGH renders the whole venue again: about 5–6 ms of GPU per camera (measured on an RTX 3060). If the FPS drop, use LIGHT, fewer outputs, 720 or FPS 30. With Resolume on the same PC, set the program to FPS 30 (main menu, GRAPHICS) and use PGM plus 1 or 2 cameras in HIGH. Always check each output FPS before recording.

In OBS: Spout needs the "Spout2 Plugin for OBS" (Off World Live) and NDI needs the DistroAV plugin; add the source and pick XRVL OUT 1 … 4. In Resolume, the outputs show up by themselves in Sources (Spout or NDI). An output name does not change when you switch cameras, so the source is never lost.

3D Editor (edit mode)

Opened with OPEN 3D EDITOR, with the space bar `Space` from the technician view (even with the menu hidden), or automatically after IMPORT XML. Everything is edited directly in Unreal, on the real screens with live video.

- The room turns translucent blue so you can see the screens from any side, and a grid appears on the floor.
- The menu hides and three areas appear: the toolbar on top, the panel to manipulate on the left (ROOM SIZE, INSPECTOR, TRANSFORM, MAGNET, PITCH and HINTS) and the panel to choose on the right (SLICES / GROUPS / ATTACHMENTS tabs).
- `Tab` hides or shows both side panels; the toolbar stays the same (with the panels hidden, switch the gizmo mode with `W` and `E`). The sections of both panels fold with the small arrow in their title. The editor always remembers how you left it (folded sections and panels), even in another session.
- To exit: `Space`, CLOSE EDITOR or `ESC`. You go back the way you came in: if the menu was showing it stays showing, and if it was hidden it stays hidden. The room returns to normal and the technician to its place. While you type in a text field, `Space` types a space and does not open or close the editor.

Editor camera (like in Unreal)

ACTION	WHAT IT DOES
Hold right button + mouse	Look around.
Right button + `W` `A` `S` `D`	Move forward, left, back and right, where you look.
Right button + `Q` / `E`	Q goes up, E goes down.
Right button + `Shift`	3 times faster while held.
Wheel	More or less flying speed.
Middle button (wheel pressed)	Pans the view.

Selecting screens

ACTION	WHAT IT DOES
Click a screen	Selects it (magenta outline). Works through walls, floor and ceiling too, and from the back of the screen.
`Ctrl` + click	Adds or removes screens from the selection, to move or rotate them together.
Click on empty space	Clears the selection.

> Selecting does not move the camera: to go to a screen use **FOCUS** in the list, or **FIT VIEW** to see them all.

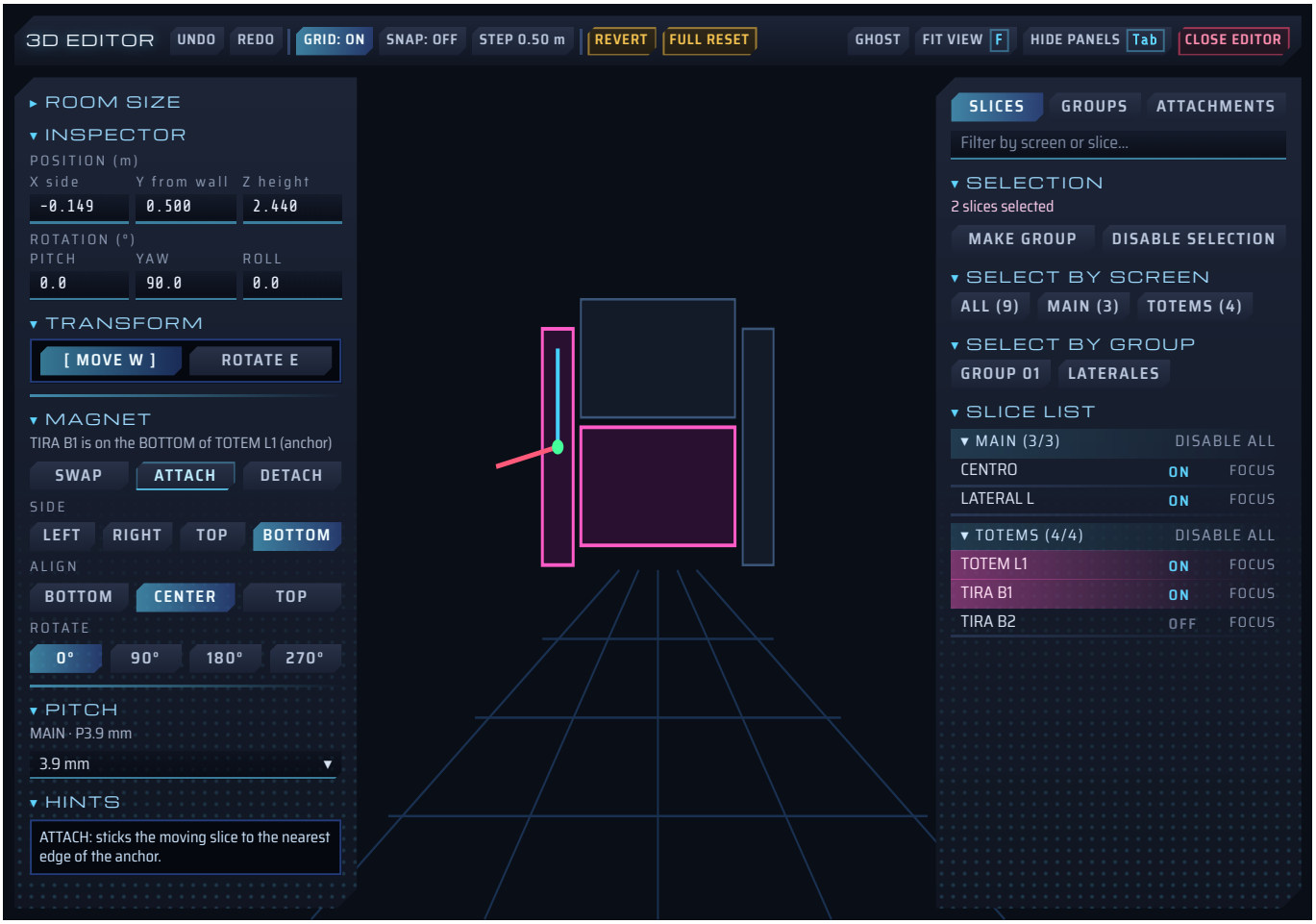
Moving and rotating with the gizmo

When you select, the gizmo appears at the center of the selection. Colors show the room axes: **red** = sides (X), **green** = depth (Y), **blue** = height (Z).

ACTION	WHAT IT DOES
`W` or **MOVE (W)**	Move mode: three arrows.
`E` or **ROTATE (E)**	Rotate mode: three rings.
Hover over the gizmo	The axis you are about to grab lights up.
Drag an arrow	Moves along that axis only. With **SNAP: ON**, it jumps in **STEP** increments.
Drag a ring	Rotates around that axis, from the center of the selection.
`Shift` (either) + ring	Rotates in 45° steps.

> With several screens selected the whole set moves or rotates. Clicking the gizmo takes priority over any screen behind it. Every finished drag is one **UNDO (Ctrl+Z)** step.

Editor toolbar and panels



3D editor: top toolbar, left panel and right panel (reference)

Top toolbar, from left to right:

- **UNDO (Ctrl+Z)** **REDO (Ctrl+Y)** — undo and redo every finished change (gizmo, inspector, ON/OFF, ENABLE/DISABLE ALL, room, magnet, REVERT and FULL RESET), up to 50 steps. The history survives closing and reopening the editor; it starts over when you load another venue or import an XML.

- GRID: ON SNAP: OFF STEP 0.50 m — GRID shows or hides the floor grid. SNAP makes positions jump in steps (in the inspector and on the gizmo arrows). STEP picks the step (0.10 / 0.25 / 0.50 / 1.00 m); the grid has a line per step and a stronger one every 1 m, starting at the center (X) and at the back wall (Y).
- REVERT — goes back to the last saved venue. FULL RESET — puts every screen back where the XML placed it. Both ask for a second click (CONFIRM?) and can be undone with UNDO.
- IMPORT XML — imports another Resolume XML without leaving the editor: it closes, loads the new venue and reopens by itself. If the venue has unsaved changes, it warns you first.
- DISABLED: GHOST / DISABLED: INVISIBLE — how disabled slices look while editing: GHOST shows them translucent (they cannot be selected) and INVISIBLE hides them. Outside the editor they are always hidden.
- FIT VIEW HIDE PANELS CLOSE EDITOR — frame everything, hide or show both panels (same as `Tab`; the button then reads SHOW PANELS) and leave the editor.

> **Editor words (as in Resolume): SCREEN** = each Resolume Screen (one output). **SLICE** = each LED screen in that Screen; in this guide a lowercase "screen" is a slice. **GROUP** = a group of slices you make yourself.

Left panel (manipulate). Each section has a small arrow in its title that folds or unfolds it, so you only keep in view what you use:

- **ROOM SIZE** — room WIDTH, LENGTH and HEIGHT, in meters. Applies with `Enter` or when leaving the field. Every IMPORT XML starts at 30 × 20 × 6.
- **INSPECTOR** — exact position and rotation (see below).
- **TRANSFORM** — MOVE (W) ROTATE (E), the gizmo mode (the active one is filled in cyan).
- **MAGNET** — joins screens by their edges (explained in its own section below).
- **PITCH** — the pitch (LED spacing) of each SCREEN, which sets the real size of its screens (explained below).
- **HINTS** (at the very bottom) — live help: hover any button (including ON, FOCUS and DISABLE ALL in the list and the ones in its title) and it explains what it does; otherwise it shows a tip based on what you have selected. Once you know the program you can fold it.

INSPECTOR — exact position (POSITION row) and rotation (ROTATION row) of the selection:

	WHAT IT DOES
X side	Meters to the sides, from the room center.
Y from wall	Meters from the back wall.
Z height	Height in meters above the floor.
PITCH / YAW / ROLL	Rotation in degrees.

Type a value or drag over the field. Holding `Shift` while dragging Pitch, Yaw or Roll rotates in 45° steps. With several screens selected, the whole set moves and rotates around its center. While you drag the gizmo, values update live (with several screens, on release).

Right panel (choose): three tabs, SLICES, GROUPS and ATTACHMENTS. **SLICES** is the main one, top to bottom:

- The field at the top filters by slice or Screen name. The SELECTION, SELECT BY SCREEN, SELECT BY GROUP and SLICE LIST sections fold with the small arrow in their title.
- **SELECTION** — how many slices are selected. MAKE GROUP makes a group from the selection: a small field appears with a suggested name (GROUP 01, GROUP 02…); type another one if you like and press `Enter`, or just press `Enter` to accept it. DISABLE SELECTION disables every selected slice at once (undo with UNDO).
- **SELECT BY SCREEN** — one button per Resolume Screen, plus ALL (n): a click selects all its slices.
- **SELECT BY GROUP** — one button per group of yours: a click selects it. If you have not made any yet, it tells you.
- **SLICE LIST** — grouped by Screen: the small arrow folds or unfolds its slices and ENABLE ALL / DISABLE ALL enables or disables the whole Screen. On the other side of the title there are three buttons, all remembered: HOVER shows the full name when the mouse rests on a row; LABELS writes each slice name on its screen (only inside the 3D editor); « widens the list toward the center, over the 3D view, just enough to read the longest name, without changing the panel (» puts it back).
- Click a row to select it (`Ctrl` + click adds or removes). The selected row turns magenta.

- ON / OFF is a button: click to disable that slice (it turns OFF) and click again to enable it. A disabled one is hidden in Unreal and can be enabled again any time.
- FOCUS brings the camera in front of that screen.

GROUPS — the list of your groups, one per row with its name clearly visible and four buttons:

- SELECT selects it. UPDATE replaces it with the current selection.
- RENAME changes its name: type the new one in the field that appears and press Enter.
- DELETE deletes it (asks for a second click: CONFIRM?). Deleting a group does not delete the screens.
- Groups are made with MAKE GROUP in the SLICES tab. They are saved with the venue (SAVE VENUE) and loaded when you open it.
- **ATTACHMENTS** — the list of magnet attachments (explained below).

PITCH: the real size of the screens

Each screen size comes from its resolution in Resolume multiplied by the LED pitch. For example, 1920 × 1080 pixels at 4 mm pitch measure 7.68 × 4.32 m. Since in a real show each SCREEN may feed a processor with different modules, the pitch is chosen per SCREEN:

- Select a screen: the PITCH section shows which SCREEN it belongs to and its pitch (with nothing selected it shows nothing).
- Pick the pitch from the dropdown, or CUSTOM to type another value in millimeters and confirm with Enter.
- The change applies to every screen of that SCREEN at once. Each screen keeps its bottom edge in place (horizontal ones, their center), so nothing sinks into the floor or floats away.
- Screens joined with the magnet rearrange themselves to stay attached.

> Each SCREEN pitch is saved with the venue (SAVE VENUE) and is part of UNDO/REDO. Old venues without a saved pitch use the default pitch.

Magnet (MAGNET): joining screens by their edges

Attaches a screen to the edge of another, like in the real rig (columns, L-shaped screens, side wings), and keeps it attached even when you move or rotate the set. Each attachment has two roles:

- **ANCHOR**: the one that stays put.
- **MOVING**: the one that sticks to the anchor.

Attaching two screens

1. Click the screen that will stay put (anchor) and Ctrl + click the one that will move. The MAGNET text confirms which is which.
2. Bring the moving screen close to the edge you want and press ATTACH: it sticks to the nearest edge of the anchor without turning. The panel highlights the side, alignment and rotation it picked.
3. If needed, adjust with **SIDE (of the anchor)** LEFT RIGHT TOP BOTTOM, **ALIGN** BOTTOM CENTER TOP (with side TOP or BOTTOM they become LEFT / CENTER / RIGHT) and **ROTATE (moving screen)** 0° 90° 180° 270°, which turns the moving screen on itself without detaching it.

With an attachment selected

BUTTON	WHAT IT DOES
SIDE / ALIGN / ROTATE	Change the attachment instantly, without detaching and attaching again.
SWAP	Swaps the roles without moving anything: the moving screen becomes the anchor (with groups, the groups swap). Without an attachment, it only swaps the order of the two selected screens.
DETACH	Removes the attachment. The screens stay where they are.

Moving attached screens

- Select only the moving screen: the gizmo sits on the shared edge. In MOVE there is a single arrow that slides it along the edge without detaching it. In ROTATE there is a single ring that folds it like a hinge (`Shift` = 45° steps).
- Typing the moving screen position in the INSPECTOR also slides it along the edge.
- If you move or rotate the anchor, its attached screens follow, in a chain.

Attaching two groups

- With the SELECT BY SCREEN or SELECT BY GROUP buttons (SLICES tab), click the group that stays put and `Ctrl` + click the group that moves. Both the Screen groups and your own work. Both get selected.
- Press **ATTACH**: the hinge is made between the two screens with the nearest edges, and the whole moving group follows it as a rigid block.
- SIDE, ALIGN, ROTATE, SWAP and DETACH work just like with two screens. With only the moving group selected, the arrow slides the block along the edge and the ring folds it.
- The two groups cannot share screens, and nothing in the moving group can already be the moving part of another attachment.

ATTACHMENTS

The **ATTACHMENTS** tab lists every attachment in the venue, one per row: **ANCHOR > MOVING (SIDE)** (group attachments show each group name). **SELECT** selects that pair to edit it (handy with many screens) and **DETACH** removes it (asks for a second click: CONFIRM?).

> A screen can be the anchor of several attachments but the moving screen of only one, and loops are not allowed. Attachments are saved with the venue (**SAVE VENUE**), are part of UNDO/REDO and are cleared by **IMPORT XML**.

Keep in mind

- Screens are not deleted in the editor: fix extra or missing ones in Resolume, or disable them (OFF) to keep them out of Unreal. Whatever is off in Resolume comes in disabled. Slices with the same name get _2, _3....
- **SAVE VENUE** saves everything: positions, room, each SCREEN pitch, groups, magnet attachments, disabled slices and each channel type (NDI or Spout, with its sender). NDI sources are not saved: they depend on each network and are chosen on each machine.
- SPOUT only works within the same PC. NDI works on the same PC and from another computer on the network.
- In the editor every screen shows a thin grey border, visible even without video (it goes away when you close the editor). Screen backs are dark grey and also clickable.
- Performance: on the same PC, Resolume and XR Venue Lab share the GPU and CPU. For big projects, run XR Venue Lab on another PC over NDI; on one PC, if it feels slow, use LOW or MEDIUM and try NDI and SPOUT.
- To share a venue: **OPEN VENUES FOLDER** and send the .json. Your colleague opens it with **IMPORT VENUE** and picks their NDI sources.